

The gap between evidence delivery and answer quality

Event memory with addressable sources in long multi-party dialogues

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Abstract

In long multi-party conversations, useful knowledge rarely coincides with a single self-contained message: states change, decisions get refined, and a late question is phrased differently from the original discussion. Raw messages preserve evidentiality but close this semantic gap poorly; free-form summaries improve access but can lose the author, the time and the per-fact link to the source. We study *event memory with addressable sources* (Elem): every derived event has its own text for retrieval and immutable pointers to the originating messages, carrying author, timestamp and a verbatim quote. Events are built before any question is read, compete with raw messages in a single ranking, and after retrieval expand back into primary evidence.

What is confirmed. The event layer reliably improves *evidence delivery*. Across all 2,400 EverMemBench questions, the precision of delivered gold sources rises by +3.21 pp (95% CI [+2.84; +3.59], $p < 10^{-59}$) while recall rises simultaneously by +0.64 pp ([+0.48; +0.80]). This is not a trade-off: events deliver *fewer* sources than raw retrieval (9.47 versus 10.00, never more), and still find *more* gold ones (+0.117 per question, [+0.083; +0.151]). The direction is positive in all nine official categories. The effect replicates on SocialMemBench (+0.90 pp precision for flat memory, $p < 10^{-31}$) and dominates query-side expansion: at statistically indistinguishable recall, events deliver +6.68 pp ([+5.06; +8.27]) denser evidence than matched HyDE.

What is not confirmed. None of these gains converts into answer quality. Across 2,400 questions micro accuracy moves by +0.38 pp (CI [-0.8; +1.6]). A registered confirmatory check on the 2,100 questions outside the discovery slice does not support the link: where events deliver strictly better evidence ($n = 650$), the accuracy gain is +0.77 pp ($p = .649$). The only nominally significant accuracy gain sits on a post hoc selected Temporal Duration slice and does not survive a Holm correction over nine categories; the endpoint whose slice and analysis were both fixed in advance (Group primary, $n = 150$) is negative.

Contribution. We report a *dissociation*: in multi-party conversational memory with a strong dense raw baseline, a derived representation measurably improves what reaches the generator, and this is not the binding constraint on answer quality. Both halves of the claim are measured on the same sealed runs, and the evidence endpoint is deterministic (0 discrepancies out of 300 under a stochastic rerun) whereas accuracy flips on 4% of questions. The practical conclusion: addressable evidence is almost free (\$1.92 per domain of ingestion against \$32.10 for Hindsight, 0.41 GB, +2 ms), but one should not expect an accuracy gain from it on current benchmarks.

1 Introduction

Conversational memory faces a conflict between two goals. Retrieval is best served by a self-contained description: “Alexey cancelled his attendance because he is leaving.” Verification needs the original material: who exactly wrote this, when, and in what words. If the history is replaced by summaries the answer becomes more accessible, but the boundary between evidence and interpretation disappears. If only messages are kept, the proof survives, but a late question has to guess anew all the elliptical, scattered and revised messages.

The problem is especially visible in a group. The author of a message, the owner of an opinion and the subject of a statement can be three different people. Several topics run at once, and positional proximity in the stream does not mean membership in the same line of conversation. Finally, states evolve: “I will be at the meeting” becomes “I have left town” a week later. A flat profile will easily either keep both versions as current, or erase the first one and deprive the system of any way to explain the change.

We propose to treat derived memory not as a new source of truth but as an *index over evidence*. In our implementation, Elem, an event contains (i) text optimized for semantic retrieval; (ii) a type, a viewpoint

WHY RAW MESSAGES ARE NOT ENOUGH

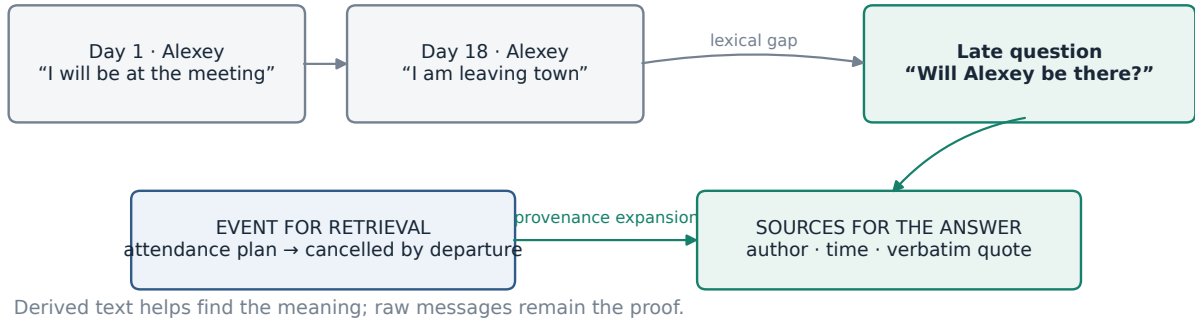


Figure 1: Two-view memory: a normalized event serves as the semantic bridge, while the answer is built from the returned primary messages. The example is illustrative and is not part of any benchmark.

owner, a subject and an observed time; (iii) an immutable set of source IDs and literal quotes. Events compete with raw turns in a single vector top- k , and the selected derived documents expand back into messages. Episodes are an optional layer: they link revision, reaffirmation and augmentation, but do not replace events or their sources.

This work answers three questions:

1. is the full system competitive against published memory systems within a consistent evaluation harness;
2. does the event representation produce an effect beyond a larger context volume, message order and strong raw retrieval;
3. does the effect transfer across different kinds of multi-party memory tasks, and does episode linking justify its additional complexity.

Contributions.

1. A two-view memory unit in which the retrieval abstraction is separated from immutable raw evidence, and author and time are preserved all the way to the answer.
2. Query-independent construction and unified retrieval over heterogeneous memory units with deterministic provenance expansion.
3. A full evaluation on EverMemBench, SocialMemBench and GroupMemBench with recorded cost, size, latency and frozen artifacts.
4. A mechanistic set of controls separating amount of text, chronological order, event representation and episode linking, including an honest negative result for linking.

2 Related work

Long-term memory for agents. Generative Agents introduce an experience stream, retrieval and reflection [Park et al., 2023]; MemGPT manages several context tiers [Packer et al., 2023]. Mem0 extracts and consolidates durable facts, and its graph variant adds relations [Chhikara et al., 2025]. Zep/Graphiti stores a temporally-aware knowledge graph [Rasmussen et al., 2025]; A-MEM organizes notes dynamically [Xu et al., 2025]; Hindsight separates facts, experience, summaries and beliefs [Latimer et al., 2025]. HippoRAG uses graph association and personalized PageRank [Gutiérrez et al., 2024].

Our goal is narrower than building universal memory: bounded multi-party chat with identifiable participants. The novelty is not in the mere use of an LLM or a vector, nor in ranking raw and derived documents jointly, which is partly anticipated by the collapsed-tree mode of RAPTOR [Sarathi et al., 2024], but in the joint construction of a *retrieval representation* + *immutable evidence address* with speaker and time attribution, and in a controlled analysis of when it helps.

Benchmarks and attribution. LoCoMo and LongMemEval evaluate multi-session recall [Maharana et al., 2024, Wu et al., 2024]. EverMemBench adds long multi-party and multi-group projects [Hu et al., 2026]; GroupMemBench controls reply structure, speaker conditioning and implicit links [Yang et al., 2026]; SocialMemBench measures facts, norms, exceptions and attribution in social groups [Owolabi, 2026]. RAG and ALCE separate retrieval usefulness from citation quality [Lewis et al., 2020, Gao et al., 2023b]. Likewise, we do not treat a valid source pointer as proof that a derived claim is true.

Derived units of indexing. The closest class of work builds a derived representation over the corpus before any query. Dense X Retrieval indexes propositions — atomic self-contained statements extracted from passages — and shows that this granularity beats passages and sentences on open-domain QA [Chen et al., 2023]. RAPTOR recursively clusters and summarizes chunks into a tree, and in its main mode (collapsed tree) flattens it into a single layer and ranks all nodes at once by cosine — both the original leaves and the generated summaries [Sarathi et al., 2024]; by their own measurements, between 18.5% and 57% of the returned nodes are not leaves.

Let us say it plainly: the idea of a single pool of raw and derived documents is not new and is partly anticipated by the collapsed tree. Our difference lies elsewhere, in the nature of the unit. Both works are methods over a document corpus: their derived units have no author and no time, and the link to the source has the granularity of a passage or a leaf, not of a per-fact address. RAPTOR itself reports that about 4% of its summaries contain small hallucinations, and its structure does not allow a specific statement to be traced back to a specific support. For multi-party correspondence this is not enough: the answer must name who said it and when, and the owner of an opinion may coincide neither with the author of the message nor with its subject. Our derived text is therefore accompanied by an immutable set of (source ID, literal quote) pairs checked by a validator, and by separate owner and subject fields.

The conclusions differ too. Dense X reports a broad gain for a fine derived unit; on multi-party chat we obtain a narrow one — events help where evidence is scattered in time, and do not help on average (§7). Plausible reasons for the difference are a different domain and a markedly stronger dense raw baseline than passage retrieval over Wikipedia.

What representations are compared against. Query rewriting and HyDE can reduce the lexical gap on the question side [Gao et al., 2023a]; event memory does this on the corpus side, once, before QA, and preserves the link to evidence. We compare both mechanisms on Temporal Duration: HyDE does not change accuracy relative to Raw either at top-10 or after matching the token budget and chronological order.

Approach	Derived	Versions	Raw prov.	Raw+mem pool	Before QA
Full context / BM25	–	–	yes	–	yes
Mem0 / summaries	yes	partial	depends	usually no	yes
Zep / Graphiti	yes	yes	depends	no	yes
Hindsight	yes	yes	trace	no	yes
RAPTOR	yes	–	leaf	yes	yes
Dense X	yes	–	passage	no	yes
Elem (ours)	yes	optional	quote	yes	yes

Table 1: A conceptual, not exhaustive comparison. “Raw provenance” is the granularity of the return path to the original source; specific implementations of the listed systems may differ. RAPTOR in collapsed-tree mode ranks summary nodes and original leaves in one pool, so a joint pool by itself is not our distinguishing feature.

3 Problem statement

Let $M = (m_1, \dots, m_n)$ be the history of a single bounded conversation space. A message $m_i = (id_i, g_i, a_i, t_i, x_i)$ contains the space, the author, the time and the text. A builder B receives only M and creates memory $\mathcal{D}$ before QA opens. For a query q , retrieval selects documents $R_k(q, \mathcal{D})$, after which the generator answers from the expanded context C_q .

We distinguish three properties:

1. **pointer integrity**: the source ID exists, and the quote is literally contained in it;
2. **semantic support**: the source text actually entails the derived description;
3. **answer correctness**: the generator used the delivered context correctly.

A mechanical check guarantees only the first. The second requires an entailment or human audit, the third is measured by benchmark scoring.

4 Method

4.1 The immutable event

An event has the form

$$e = (z, o, s, k, r, \tau, P, L),$$

where z is the self-contained index text, o the viewpoint owner, s the subject, k the event type, r the temporal mode, τ the observed time, P the evidence and L the local context. Evidence consists of pairs $(id_i, quote_i)$; time is derived from the real timestamps of the sources.

Separating o from s prevents naive attribution collapse: “A called B unreliable” is stored as A’s opinion about B, not as an unconditional fact about B. The types cover state, opinion, decision, commitment, relation, exception and stable pattern. A conversational reaction with no long-term content may produce no event; a cap of at most two events per turn applies.

4.2 Query-independent extraction and validation

The history is split only along natural benchmark session boundaries: in EverMemBench by day and group; in GroupMemBench by channel/phase with blocks of up to 40 messages. Forced structured output permits an empty list. A deterministic validator then removes invented IDs, missing quotes, content-free values and invalid enums. Local context may help resolve a pronoun, but does not automatically become evidence.

4.3 Optional versioned episodes

An episode $E_j = ((e_1, \rho_1), \dots, (e_t, \rho_t))$ is an append-only line where $\rho \in \{\text{create, revise, reaffirm, augment}\}$. For a new event, candidates are searched only within the conversation network. The score is

$$S(e, E) = 0.6 \cos(\phi(z_e), \phi(A_E)) + 0.4 \cos(\phi(z_e), \phi(z_{last}(E))).$$

The three best episodes go to an LLM resolver; when confidence is insufficient a new episode is created. The index keeps a hot view of the last five events, but cold events and evidence are not deleted. A false split is cheaper here than a false merge: split events can still enter retrieval independently, whereas an erroneous merge systematically brings in extraneous sources.

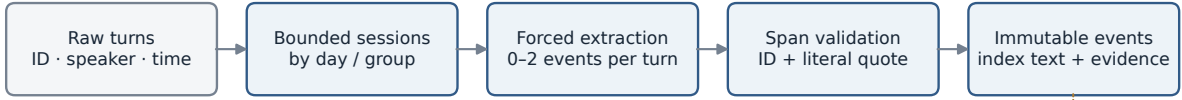
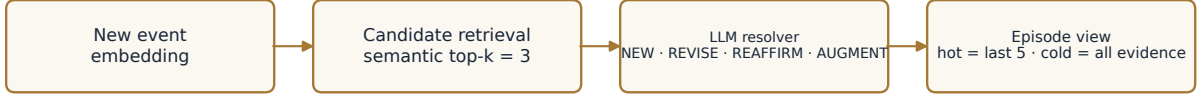
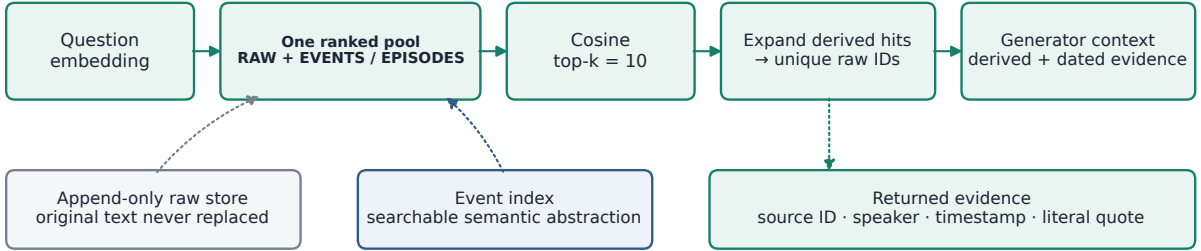
4.4 Unified retrieval and provenance expansion

A raw turn and a derived document share a common interface

$$d = (id, kind, index_text, render_text, source_ids).$$

The question vector is compared against a single pool; the top-10 may contain both kinds. After ranking, each event/episode expands into unique raw turns. The generator sees the derived text as a navigational hint and the dated, speaker-grounded sources as the basis of the answer.

This differs from “show all the profiles”: derived memory is filtered by the same relevance criterion as raw history. Static profiles and a description of the personal chat are not used in the experiments.

A • QUERY-INDEPENDENT INGESTION**B • OPTIONAL TEMPORAL LINKING****C • UNIFIED RETRIEVAL + PROVENANCE EXPANSION**

INVARIANT • the derived representation helps find meaning; raw evidence stays addressable and immutable.

Figure 2: The full architecture. Memory is built query-independently and sealed before QA. In the main variant raw turns and episodes are ranked jointly; in the ablation, episodes are replaced by separate events. A derived hit always expands into raw sources.

5 Experimental protocol

5.1 Data and conditions

Benchmark	QA	Spaces	Main test
EverMemBench	2,400	5 projects	9 task types; temporal controls
SocialMemBench	1,031	43 networks	social attribution, norms, updates
GroupMemBench	745	4 domains	reply structure, implicit, multi-hop

Table 2: External datasets used. For GroupMemBench, 102 initially empty generation outputs were restored by a separate sealed amendment; the original protocol result is retained.

The main paired baseline is a strong dense Raw, not a weak keyword-only search. The Events and Episodes conditions add derived documents to the same top-10. For Temporal Duration we additionally fix the token count, the set of expanded raw sources and the chronological order; HyDE generates a hypothetical document only for the query embedding and returns the same raw documents. All variants use identical official answer/judge prompts inside the corresponding benchmark adapter.

EverMemBench uses GPT-4.1-mini for answering, Gemini-3-Flash-preview as the open-answer judge, text-embedding-3-large for retrieval and gpt-4o-mini-2024-07-18 for memory construction. The answer and judge models, their prompts and the exact multiple-choice scoring are taken from the official harness; providers are pinned (OpenAI and Google AI Studio respectively) with fallbacks disabled — that is, our rows and the published ones do not differ by the judge model. All calls run at temperature = 0. SocialMemBench follows the GPT-4o-mini official-compatible harness. GroupMemBench uses the GPT-5 of the official run; the provider’s model version was not pinned by date, so byte-level reproduction of these numbers is not guaranteed. Comparisons across benchmarks are not to be read as comparisons of models.

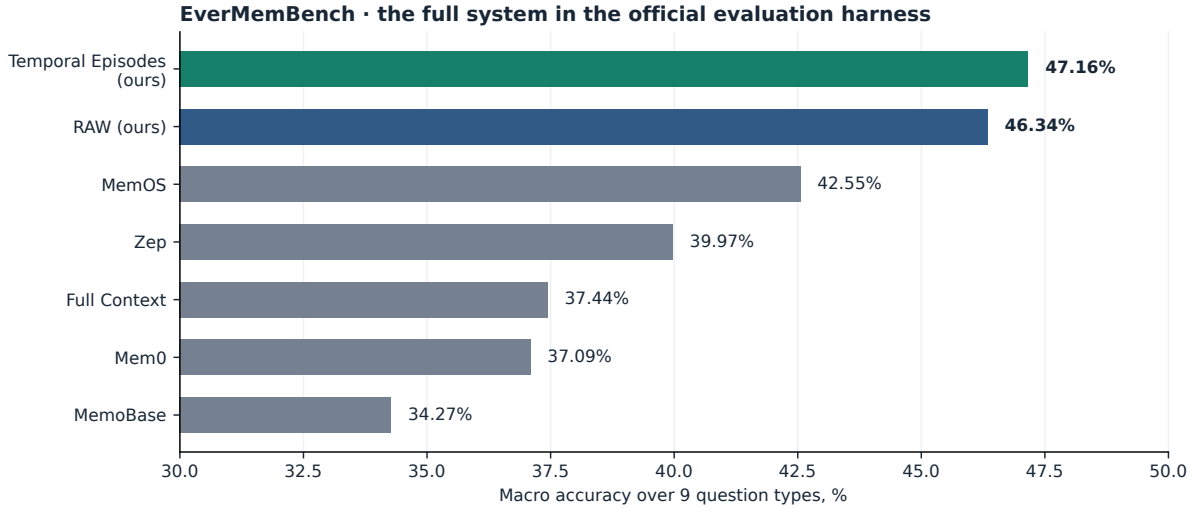


Figure 3: Macro accuracy over the nine EverMemBench categories with the GPT-4.1-mini answer backbone. Published results are taken from the benchmark paper [Hu et al., 2026]; our rows come from an adapter evaluation on top of the official harness. This is not an official leaderboard submission and not a paired significance test against other systems.

5.2 Metrics and statistics

We report micro accuracy, macro accuracy over question types, evidence recall/precision from the intersection of expanded source IDs with gold anchors — the reference spans released by the benchmark, that is, the messages marked as the support of the answer — paired differences, question bootstrap 95% CI and exact McNemar. For the five Ever projects a cluster/sign-flip analysis is computed in addition. Evidence metrics are not entailment and not the quality of user-facing citations.

Memory and inputs are stored in JSONL/SQLite, sealed with SHA-256 and reused across controls. Temporal Duration was selected after inspecting the full breakdown and is marked exploratory everywhere. The family of multiple comparisons for the mechanistic claim is the nine official EverMemBench categories; we report the Holm correction with $m = 9$ next to the nominal p values for it (§7). A separate power ceiling applies: with five project clusters the minimum attainable two-sided sign-flip value is $2/2^5 = .0625$, so cluster-level confirmation is unreachable on these data under any outcome.

Two endpoints and their resolving power. We report results at two levels, and they differ not in importance but in measurability. *Accuracy* passes through the generator and the judge; *evidence recall/precision* are computed mechanically from expanded source IDs and gold anchors and depend on neither. The difference is visible empirically: on a repeated run of one and the same condition over the same 300 questions, accuracy produced 12 flipped verdicts out of 300 (4.0%; 6 in each direction, a net shift of exactly 0.00 pp), whereas the evidence metrics agreed *on all 300 questions*. Accuracy is a noisy proxy with ceilings on easy categories; the evidence endpoint is deterministic and resolves effects that accuracy at the available n cannot resolve in principle. We therefore present evidence as the main measured result and accuracy as the question of whether it converts (§7.3), and deliberately do not substitute one for the other.

6 Results: the system as a whole

6.1 EverMemBench: competitiveness

The full Episodes system reaches **47.16% macro**, and Raw 46.34%. Our adapter result is above the values reported by Hu et al. [2026] with the same answer backbone: MemOS 42.55%, Zep 39.97%, Mem0 37.09%, Full Context 37.44% and MemoBase 34.27%. The gap to the best published memory system is 4.61 pp.

What this means. Even with imperfect linking, the full system shows a high absolute level: our adapter result is above every GPT-4.1-mini memory row published by Hu et al. [2026]. This is not a controlled rerun of other systems and not an official leaderboard submission. However, our own strong Raw already exceeds them; therefore all 4.61 pp cannot be attributed to event/episode memory.

Condition	Micro	Macro	Evid. recall	Evid. precision
Raw	49.46	46.34	8.08	21.29
Events	49.83	46.82	8.71	24.50
Episodes	50.21	47.16	10.75	22.45

Table 3: EverMemBench, all 2,400 QA, percent. Macro is the mean over the nine official question types; micro is over individual QA.

At the micro level Episodes improve 49.46%→50.21%: +0.75 pp, 95% CI $[-0.5; +2.0]$, $p = .256$. Events give 49.83%: +0.38 pp, CI $[-0.8; +1.6]$, $p = .589$. The overall gain within our paired comparison is therefore not statistically resolved. This does not contradict the high absolute macro: the questions “how strong is the system?” and “how much of the effect does the component add?” are different.

6.2 Transfer across benchmarks

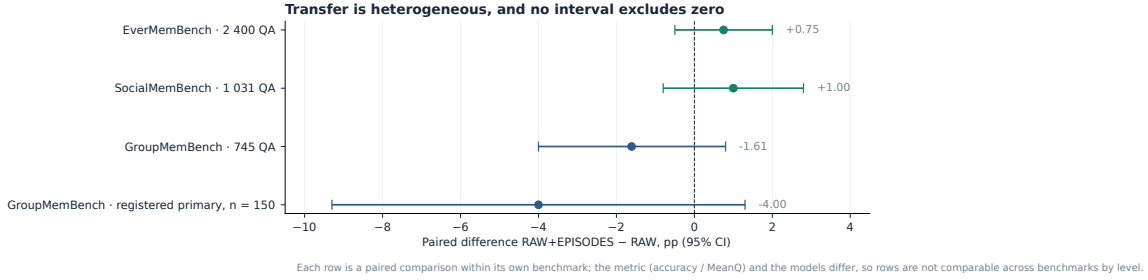
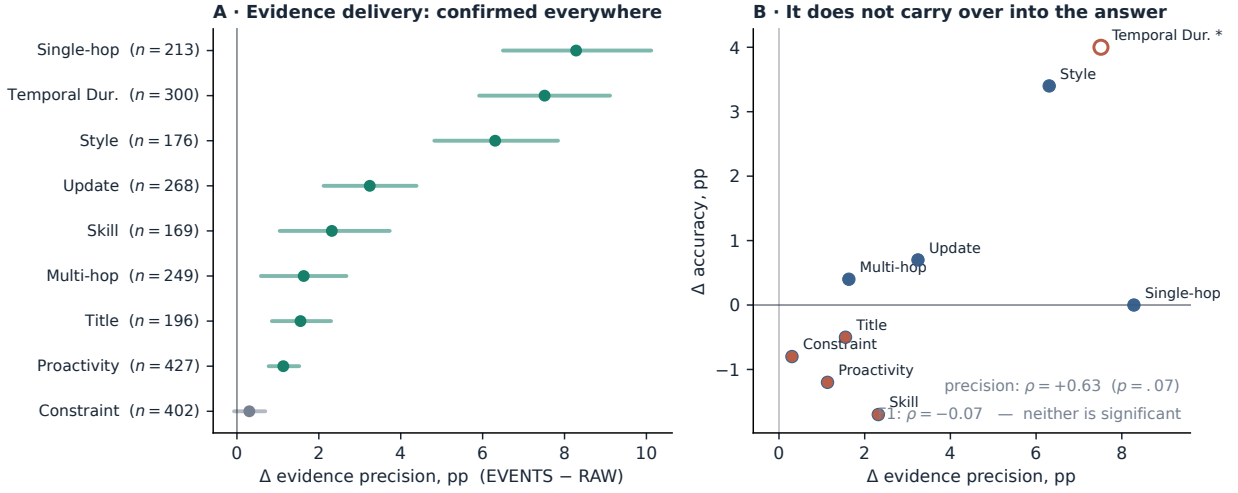


Figure 4: Paired differences Episodes–Raw within each dataset with question- or network-bootstrap 95% CI. Absolute levels are deliberately not shown: the metric (accuracy versus MeanQ), the protocols and the answer models differ across the three datasets, and comparing them by bar height would be incorrect. Group is shown after the targeted repair of empty generations; its separately registered temporal/update endpoint ($n = 150$ of 745) is shown as its own row. No interval excludes zero.

SocialMemBench gives MeanQ .358 for Raw and .368 for Episodes: +0.010, network-bootstrap 95% CI $[-.008; +.028]$. The result is practically identical to the published full-context .369 and above the listed benchmark memory rows (best Subject-Mem .321), but there is no significant gain of its own.

On accuracy, Social is thus a draw. But at the level of evidence delivery the same run gives a resolved positive effect, and this is the only cross-benchmark transfer the work managed to establish at all. Paired over 1,031 questions against Raw: flat memory +0.90 pp precision (CI $[+0.70; +1.11]$, $p < 10^{-31}$), versioned +0.86 pp ($[+0.65; +1.06]$), episodes +0.34 pp ($[+0.17; +0.51]$) at +0.99 pp recall ($[+0.16; +1.82]$). The magnitude is markedly smaller than on Ever (+3.21 pp), and the reason is transparent: the base evidence recall on Social is 67.5% against 8.1% on Ever, so there is almost nothing left to improve. The direction, however, reproduces on a second dataset, with a different model and a different harness.

On the pre-registered Group primary (Finance+Technology, knowledge update+temporal, $n = 150$) the original protocol result was .340→.300. A targeted amendment repeated only the 102 empty generations with the same sealed prompts and a larger output budget; 102/102 were restored, and the remaining 1,388 answers were left unchanged. After the repair Raw reaches .373 and Episodes .333: the delta stays at −4.0 pp, CI $[-9.3; +1.3]$, $p = .210$. Over all 745 QA the result is .431→.415, −1.6 pp, CI $[-4.0; +0.8]$, $p = .246$. The negative direction is therefore not an artifact of the empty outputs. The repair cost \$2.69; 94% of its output tokens went to hidden reasoning, which confirms the cause of the original failure.



Left: paired difference over 2,400 questions, question bootstrap 95% CI; in grey, the only category whose interval covers zero.
Right: the same X axis against the shift in accuracy. The category with the largest delivery gain (Single-hop) sits at the accuracy ceiling and gains nothing.

Figure 5: The dissociation. **A:** paired difference in evidence precision Events—Raw over the nine official categories, all 2,400 questions, question bootstrap 95% CI; the sign is positive everywhere, the interval covers zero only for Constraint. **B:** the same axis against the shift in accuracy; the asterisk marks the post hoc discovery slice. The category with the largest delivery gain (Single-hop, +8.28 pp) sits at an accuracy ceiling of 89.7% and gains nothing. The sign of the categorical correlation depends on the choice of evidence metric and at $n = 9$ is not resolved in either direction (§7.3); the conclusion of this work rests on the stratified test over 2,100 questions, not on these nine points.

It matters that both the slice and the analysis were fixed in advance here: the Finance+Technology domains, the knowledge update+temporal types, $n = 150$ and the statistics were written into the protocol before the run. Unlike the temporal effect (§7), where the advance fixing covers only the endpoint and not the choice of slice, this result is protected from selection on both sides — and it is negative. The asymmetry of status between the negative and the positive results of this work does not point in our favour, and we consider it necessary to say so directly.

Boundary of generalization. The method is not a universal replacement for raw retrieval. On Social it is neutral on average; on Group the current version may displace useful speaker-conditioned raw turns even after the repair. This negative transfer is part of the result, not a hidden failure.

7 The event representation: what it changes and what it does not

This section is split by endpoint. First we show the effect that is measured directly and confirmed on the whole dataset: events change which evidence reaches the generator (§7.1). Then, where and how this is reflected in accuracy (§7.2), and finally the registered check of whether the two are linked at all (§7.3). To anticipate: the first is confirmed, the third is not.

7.1 Evidence delivery: the confirmed effect

We compare, pairwise and per question, the share of delivered gold sources (recall) and their density in the context (precision) over all 2,400 questions — with no slice selection.

Events—Raw, paired: precision +3.21 pp (95% CI [+2.84; +3.59], Wilcoxon $p = 2.9 \cdot 10^{-60}$, 855 questions better against 255 worse), recall +0.64 pp ([+0.48; +0.80], $p = 4.6 \cdot 10^{-13}$). The sign is positive in all nine official categories, and the CI excludes zero in eight out of nine (appendix, Table 9).

This is not a mechanical artifact of volume. Precision could have risen on its own if events simply delivered less text. The check gives the opposite: Events deliver fewer sources than Raw (9.47 versus 10.00)

Condition	Sources	Gold hits	Evid. precision
Raw	10.00	2.129	21.29
Events	9.47	2.245	24.50
Episodes	12.38	2.795	22.45

Table 4: EverMemBench, all 2,400 QA: how many unique sources entered the context, how many of them turned out to be gold anchors, and what share of the context was gold. Events dominate Raw on both coordinates at once.

and *never* more — while finding more gold ones: 2.245 against 2.129 hits per question, a delta of +0.117 (CI [+0.083; +0.151]). Less context, more evidence — dominance, not a trade-off; on 449 questions it is strict. Episodes behave differently: they buy recall with volume (+2.68 pp recall at +2.4 sources), and their precision grows more weakly (+1.16 pp). This is the first place in the work where linking and separate events differ measurably — but they differ in cost, not in usefulness for the answer (§7.4).

Query-side expansion does not reproduce this. Matched HyDE on Temporal Duration, at aligned budget and order, gives recall statistically indistinguishable from Events (−1.01 pp, CI [−2.20; +0.17]), but a markedly sparser context: Events–HyDE in precision is +6.68 pp (CI [+5.06; +8.27], $p = 9.3 \cdot 10^{-16}$). HyDE is not useless in itself — against token-matched Raw it gives +1.18 pp precision and +1.61 pp recall — it is dominated. Expansion on the query side and normalization on the corpus side move different coordinates.

What is established. The derived layer improves evidence delivery reliably, over the whole dataset, in every category, with replication on a second benchmark and dominance over query-side expansion. This is measured by a deterministic metric that depends neither on the generator, nor on the judge, nor on the choice of slice. *Whether this gain remains useful* is a separate question, and the answer to it is negative (§7.3).

7.2 Accuracy: where the effect is visible and why this is the weak spot

We analyse Temporal Duration as the slice with the largest *relative* gain of the full Ever breakdown: Raw 12.3%, Episodes 20.0%, that is +7.7 pp and +63% over the base (Table 8). In absolute points, a slightly larger shift is given by Style (30.1% → 38.1%, +8.0 pp), but at a three times higher base and with no hypothesis about a mechanism; for Events without controls, Temporal Duration is the largest in absolute points as well (+4.0 pp against +3.4 pp for Style). We do not explain the gain on Style and do not include it in the claims of this work (§10). The choice of slice itself remains post hoc. But derived documents expand into several messages, so the generator sees 17.4 sources on average instead of 10. To separate representation from volume, we gave Raw the same token budget; then we separately aligned chronological order and replaced linked episodes with independent events. A matched HyDE control tests whether query-side expansion can produce the same effect without building event memory.

Condition	Accuracy	Evid. recall	Evid. precision
RAW top-10	12.3	12.7	29.6
HyDE-RAW top-10	12.0	14.1	32.4
RAW token-matched	12.3	19.1	21.5
RAW token-matched chrono	14.0	19.1	21.5
HyDE-RAW token-matched chrono	14.0	20.7	22.6
EVENTS token-matched	17.0	19.7	29.3
EVENTS token-matched chrono	19.3	19.7	29.3
EPISODES ranked	20.0	20.5	31.1
EPISODES chrono	18.3	20.5	31.1

Table 5: Temporal Duration, $n = 300$, percent.

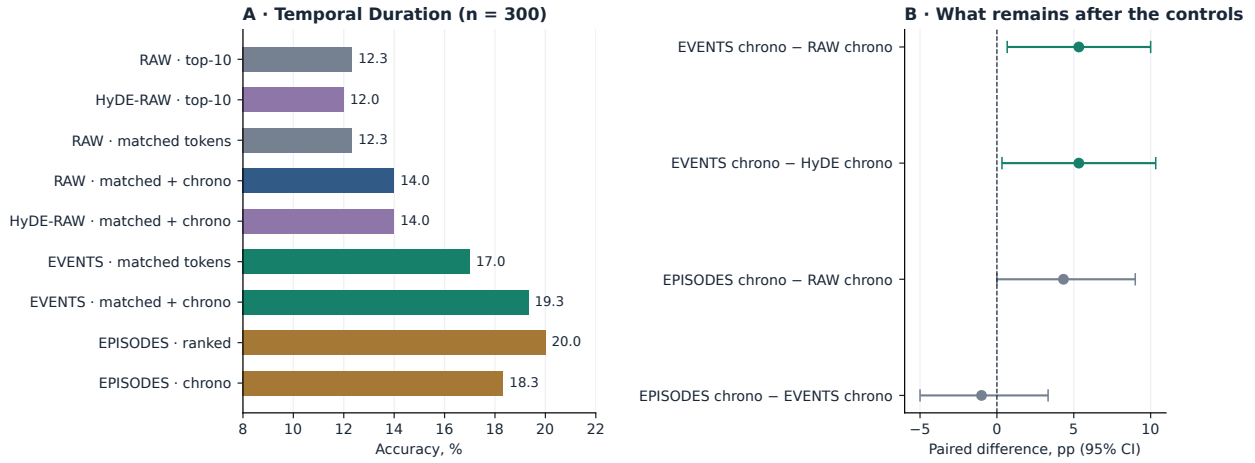


Figure 6: Mechanistic controls on the exploratory Temporal Duration slice. EVENTS chrono retain an advantage over RAW and HyDE at equal budget and order; EPISODES do not exceed EVENTS. Intervals are question bootstrap 95%.

At equal token budget and order, Events reach 19.3% against 14.0% for Raw: +5.33 pp, 95% CI [+0.67; +10.0], nominal $p = .0402$. Matched HyDE also gives 14.0%; the difference Events–HyDE is +5.33 pp, CI [+0.33; +10.33], nominal $p = .0479$ (37 wins against 21 losses). The direction is positive in all five projects, but with five clusters the minimum attainable sign-flip is $p = .0625$. Over all 2,100 questions outside Temporal Duration, Events give -0.14 pp, CI $[-1.33; +1.10]$, $p = .877$. The effect is not explained by a simple “derived memory is better everywhere”; it concentrates where scattered temporal boundaries have to be found.

The mechanistic picture does not reduce to a single retrieval metric. Raw top-10 already has 29.6% precision of expanded gold sources — no lower than Events (29.3%) — but at 12.7% recall it yields only 12.3% accuracy. To raise recall, raw retrieval has to widen the window: token-matched Raw reaches 19.1% recall, precision falls to 21.5%, and accuracy grows only to 14.0%. HyDE shifts the same curve further into recall (20.7%) at 22.6% precision — and accuracy does not change at all. Events hold both coordinates at once: recall 19.7% at the level of token-matched Raw and precision 29.3% at the level of Raw top-10, which is what 19.3% corresponds to.

On this slice events hold both coordinates at once, whereas raw retrieval gets them one at a time: two conditions with almost the same recall (19.1% and 19.7%) give 14.0% and 19.3%, and two conditions with almost the same precision (29.6% and 29.3%) give 12.3% and 19.3%. HyDE fits the same picture: it moves recall only and does not reproduce the event gain.

An explanation suggests itself — “accuracy follows the recall–precision frontier of delivered evidence”. We tested it and did not confirm it (§7.3); the explanation itself remains a description of four rows of Table 5, not a mechanism. This is still a post hoc discovery slice over five project clusters; the project-level sign-flip is $p = .1875$, and 288/300 verdicts agreed in a stochastic rerun.

Correction for multiplicity. Temporal Duration was selected from nine official categories after inspecting the full breakdown, so the family is $m = 9$. The first Holm step compares the smallest p of the family against $\alpha/9 = .0056$; our .0402 and .0479 exceed it, so the procedure stops and no comparison survives the correction (Bonferroni equivalents .36 and .43). The remaining eight p values are not needed for this conclusion. Together with the power ceiling (.0625 at five clusters, §5.2) this means the effect has no confirmatory statistical status either at the question level or at the project level.

What exactly was pre-registered. The endpoint of the matched-HyDE comparison was fixed before any outcomes were obtained, but the slice itself was not. Fixing the analysis in advance does not protect against selection of the slice, so registration here covers only the second half of the procedure. We read the temporal result as hypothesis-generating: it is stated narrowly enough for an independent confirmatory replication to refute it, and we do not present it as a confirmed effect.

7.3 Does the gain convert into the answer: a registered check

An explanation that only ever sees the slice on which it was found cannot turn out to be wrong. We therefore registered a prediction before the computation: if the frontier is a mechanism, it must also describe the 2,100 questions where events do not help. The protocol, the reading rule and the power ceilings were fixed in advance; the check is fully offline and uses per-question fields of already sealed runs, with no new generations.

Each question outside Temporal Duration is assigned to one stratum by how the delivered evidence changed in the transition Raw→Events: **A** — events dominate (recall and precision no lower, at least one strictly higher), **B** — Raw dominates, **C** — a mixed case or equality. The prediction: the accuracy delta is positive in A, negative in B and indistinguishable from zero in C.

Stratum	<i>n</i>	Δacc	95% CI	Wins/losses	McNemar
A (events dominate)	650	+0.77	[−1.85; +3.54]	41/36	.649
B (Raw dominates)	209	−2.39	[−6.70; +2.39]	9/14	.405
C (mixed)	1,241	−0.24	[−1.53; +1.05]	32/35	.807

Table 6: Confirmatory check on the 2,100 questions outside the discovery slice, Events–Raw, percent and pp.

The prediction was not confirmed. In stratum A, where events deliver strictly better evidence, the gain is only +0.77 pp at $p = .649$. The direction in B is negative, as predicted, and C does indeed contain zero, but the main condition of the reading rule is not met. At the category level, the sign of the delta agrees with the direction of dominance in 1 case out of 5 ($p = .375$).

The categorical correlation, unlike the stratification, turned out to be *metric-dependent*, and we report this as it is. The correlation between the gain in evidence F1 and the gain in accuracy over eight categories is $\rho = -0.07$ ($p = .87$), but the same correlation computed on the gain in evidence *precision* is $\rho = +0.63$ over nine categories ($p = .07$) and $\rho = +0.57$ over eight without the discovery slice ($p = .14$). At $n = 9$ the critical value is 0.68, so neither quantity is resolved, and the choice between them cannot be made from the data: F1 mixes recall and precision, which move differently for Events and Episodes (§7.1). We therefore do not claim that the link is absent at the category level — we claim that it is not resolved on nine points, whereas the pre-registered stratified test on 2,100 questions does not confirm it. The conclusion of this work rests on the second, not the first.

What the check does *not* show. There are few discordant pairs — 77 in stratum A, 23 in B, 67 in C — because a change of context rarely flips an answer. At 77 pairs the minimum detectable effect in A is ≈ 2.9 pp, so the result excludes a large effect but does not resolve a small one. This is an absence of confirmation, not a demonstrated zero.

The conclusion is taken by the rule written down in advance: the frontier remains a description of the discovery slice and loses the status of an explanation. An improvement in delivered evidence does not by itself imply an improvement in the answer, and why events win specifically on Temporal Duration remains unknown.

Together with §7.1 this gives the main result of this work, and it is two-sided. The gain in evidence delivery is real, large by statistical standards and reproducible; there is no gain in the answer, and the link between them is not confirmed by a direct check. We call this a *dissociation*: in multi-party conversational memory with a strong dense raw baseline, retrieval quality stops being the binding constraint long before it becomes good. Both halves are measured on the same sealed runs, so the asymmetry of status here is not rhetorical: what is confirmed is what is easier to measure, not what is more convenient for the method.

7.4 What episodes add over events

New symmetric controls remove the advantage of linking: Episodes chrono 18.3%, Events chrono 19.3%; the difference is -1.0 pp, CI [−5.0; +3.33], $p = .755$. In the previously sealed comparison of ranked episodes 20.0% against events chrono 19.3%, the difference was only +0.67 pp, $p = .888$.

We separately note an inconsistency that we do not explain: chronological order helps Raw (+1.7 pp) and Events (+2.3 pp), but for Episodes it is harmful (20.0 → 18.3). All three shifts are single-digit question

counts at $n = 300$, and we do not treat them as distinguishable; this is more an estimate of the scale of noise than three different effects.

This is an important negative result. A versioned chain is useful for inspection and can connect specific changes, but its average contribution to retrieval is not separable from a set of individual events. Good retrieval can survive a false split by delivering both records independently, so complicating the linker is not obliged to improve the endpoint.

8 Efficiency and engineering properties

8.1 Cost and latency

The full EverMemBench experiment recorded 26,413 API calls and an upper list-price estimate of \$8.88. Of this, memory construction with gpt-4o-mini costs \$4.76, embeddings \$0.46, answers \$3.12 and the judge \$0.55. These figures refer to a research run with caches and do not equal a production TCO.

The serialized episode snapshot occupies 12.77 MB, but a full operational run together with the embedding cache is about 2.2 GB; calling the whole system “12.8 MB” is therefore incorrect.¹

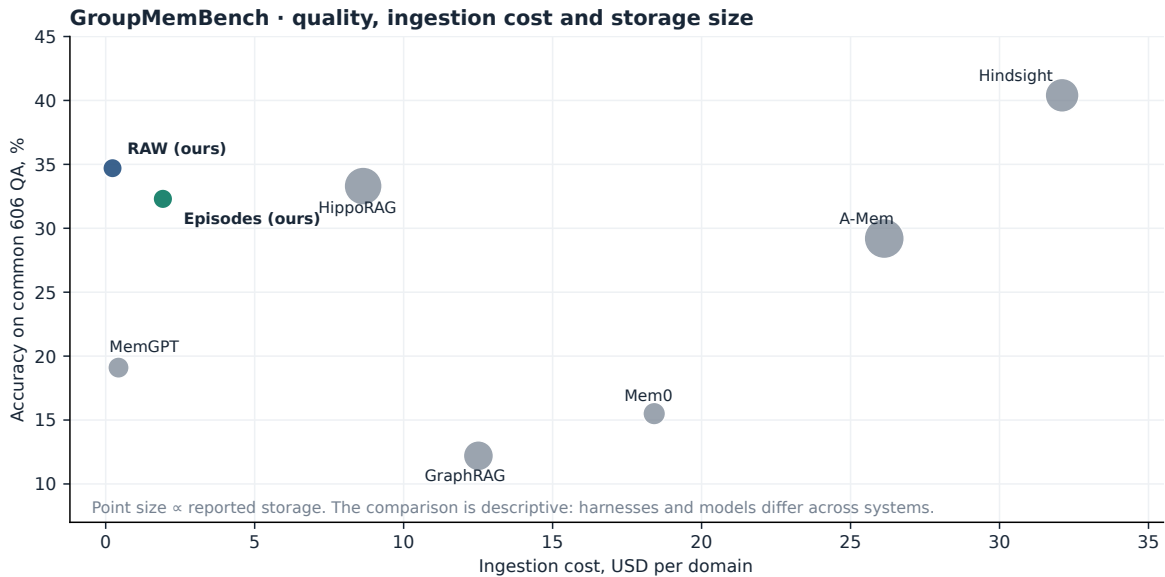


Figure 7: A descriptive GroupMemBench efficiency landscape on the shared sample of 606 QA, from the values of the benchmark paper and of our adapter. Point size encodes reported storage. Our points use the repaired amendment; cost definitions and backends differ, so the plot shows an engineering trade-off but does not support a claim of “cheaper at the same quality”.

On this shared sample Hindsight reaches 40.4% at a reported ingestion of \$32.10/domain and 4.25 GB; HippoRAG 33.3%, \$8.64 and 5.76 GB; our Raw 34.7%, \$0.23 and 0.37 GB; Episodes 32.3%, \$1.92 and 0.41 GB. The system is economical relative to heavy structural solutions, but Hindsight is more accurate. Because of the differing harness components this is context, not a controlled causal estimate of cost.

8.2 Reproducibility as part of the method

Memory construction finishes before any QA is read; snapshots and input manifests are sealed with SHA-256. Model calls are cached by content and by prompt/schema version. On interruption the process continues from the last completed unit. Derived records pass a repeated source-ID check; experiments A/B2 were run over one frozen memory without re-extraction.

¹Mean local retrieval latency grows 11.59→13.70 ms (p95 12.70→15.04 ms); this is ranking only, without network, extraction, generator and judge.

System	Accuracy, %	Ingestion, \$/domain	Storage, GB
Hindsight	40.4	32.10	4.25
Raw (ours)	34.7	0.23	0.37
HippoRAG	33.3	8.64	5.76
Episodes (ours)	32.3	1.92	0.41
A-Mem	29.2	26.13	6.60
MemGPT	19.1	0.43	0.69
Mem0	15.5	18.41	0.99
GraphRAG	12.2	12.51	3.00

Table 7: Explicit values for Figure 7. Accuracy is measured on the shared 606 QA of GroupMemBench; ingestion cost and storage are taken from the corresponding benchmark reports. Our accuracies are recomputed after the targeted repair; the cost of the repair answers is not included in ingestion. Backends, harnesses and storage definitions differ, so the comparison is descriptive.

These details do not improve the benchmark score directly, but they protect the main claim from oracle leakage and make it possible to distinguish a change of method from repeated sampling.

9 Error analysis

Extraction misses. If no event is created, the system reduces to Raw. Forced structured output substantially raises recall relative to voluntary tool calling, but does not guarantee completeness.

Attribution and semantic overreach. A literally present quote may support only part of a description; a paraphrase may mistake the author of a message for the owner of an opinion. Therefore 100% quote validity means integrity of the address, not 100% correct facts.

False split and false merge. Free-form subject/facet wording splits one line into several episodes. Retrieval often survives a split by finding both records. A merge is more dangerous: it brings irrelevant context into every future request. A conservative resolver naturally prefers NEW, which shortens the chains but protects precision.

Context displacement. A derived hit occupies a slot in the top- k and sometimes displaces a useful raw turn. This is a plausible explanation of the neutral/negative transfer outside the temporal slice and especially in the speaker-conditioned GroupMemBench.

10 Limitations

The main mechanism result was found post hoc; all of its controls use the same 300 questions and are not an independent replication. After the Holm correction over the nine official categories it is not significant ($\alpha/9 = .0056$ against the nominal .0402 and .0479), and the five Ever projects do not permit cluster-level significance under any outcome (minimum attainable $p = .0625$). Meanwhile the endpoint whose slice and analysis were both fixed in advance (Group primary) is negative. The positive conclusion of this work therefore remains hypothesis-generating and requires a confirmatory replication on an independent slice. Overall accuracy stays around 50%, and temporal below 20%.

Let us bound the confirmed result symmetrically as well. The evidence metric measures *delivery*, not usefulness: a gold anchor in the context does not mean the generator used it, and a rise in precision does not mean the derived text is semantically faithful. The absolute magnitude is modest — +0.117 gold hits per question — and a base recall of 8.1% means that more than 90% of gold evidence is retrieved by none of the conditions; we are moving a small share of a small share. Finally, gold anchors are released by the benchmark, and all conclusions inherit their completeness.

Let us bound the null separately. Memory is built by a cheap extractor (gpt-4o-mini), so the conclusion “there is no universal event gain” holds for this extraction quality, and the upper bound under a strong extractor is not measured. The cap of “at most two events per turn” and the weights 0.6/0.4 in $S(e, E)$ were chosen a priori and are not ablated. The frontier explanation does not survive its own confirmatory check (§7.3), so the observed advantage on Temporal Duration remains without a mechanism.

The source pointer is mechanically valid, but a full human entailment audit has not been completed. Review sheets exist, but are not fully annotated and are not used as a published metric. The HyDE control is complete, but uses the same 300 exploratory questions and is not an independent replication. The targeted GroupMemBench amendment restored all 102 initially empty outputs without changing the remaining 1,388 answers and verdicts. It remains a post-protocol analysis, not an official leaderboard submission. While filling the repair cache in parallel, one shared payload was called twice because of a missing inter-process lock; the extracted final answers coincided, and the duplicate changed no metric. Costs depend on prices as of 12 September 2026, on caches and on the chosen APIs; latency was measured locally.

Evidence metrics were not computed for GroupMemBench: the runner stored only the verdict. This is the only dataset with a stable negative transfer, so it is exactly where the dissociation should have been tested a third time, and exactly where we did not test it. The computation is offline and requires no new generations — the expanded source IDs for both conditions are already sealed — but at the time of writing it has not been performed, and the displacement hypothesis from §9 remains untested.

Finally, the method does not propose a new foundation model and does not solve open-domain or multimodal memory. It assumes a bounded conversation network and stable participant identifiers; upstream transcription and identity resolution are outside the central evaluation.

11 Conclusion

We investigated a simple invariant: derived memory should help retrieval without substituting itself for the proof. Elem stores a self-contained event representation together with immutable addresses of raw turns, ranks raw and derived documents in one pool, and expands retrieved memory back into speaker- and time-grounded evidence.

The first half of the invariant holds, and it is measured more reliably than anything else in this work. Events improve evidence delivery over all 2,400 questions (+3.21 pp precision at +0.64 pp recall), while delivering less text, not more; the effect is positive in all nine categories, reproduces on SocialMemBench and dominates matched HyDE. This is measured by a deterministic metric that depends neither on the generator nor on the judge.

The second half does not hold. None of these gains carries over into answer quality: across all questions accuracy does not move, and a registered check on the 2,100 questions outside the discovery slice does not confirm a link between better evidence and better answers. The only nominally significant accuracy gain was found post hoc and does not survive correction; the endpoint whose slice and analysis were both fixed in advance is negative. Social transfers neutrally, Group negatively even after the targeted repair.

We consider this dissociation the main result, not a failure. It means that in multi-party conversational memory with a strong dense raw baseline, improving retrieval stops paying off before retrieval becomes good: at a base recall of 8% the bottleneck is no longer there. The practical consequence for an engineer is direct — *store the evidence first, then build a retrieval abstraction over it*: addressability is almost free (\$1.92 per domain, 0.41 GB, +2 ms) and survives errors in the derived layer, but an accuracy gain from it should not be planned for on today’s benchmarks. The consequence for a researcher is about what to measure: accuracy at these sample sizes does not resolve effects that the evidence endpoint resolves by tens of orders of magnitude, and work that reports on accuracy alone is likely to see neither its effect nor the absence of one.

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A Full EverMemBench breakdown

Type	n	RAW	EVENTS	EPISODES
Constraint	402	79.9	79.1	79.6
Multi-hop	249	12.9	13.3	13.3
Proactivity	427	65.8	64.6	64.2
Single-hop	213	89.7	89.7	87.8
Skill	169	33.7	32.0	30.8
Style	176	30.1	33.5	38.1
Temporal Duration	300	12.3	16.3	20.0
Title	196	46.4	45.9	43.4
Update	268	46.3	47.0	47.4

Table 8: Accuracy over all official categories, percent. EVENTS here is the full experiment A without token/order controls.

B Evidence delivery by category

Type	n	Δ precision	95% CI
Single-hop	213	+8.28	[+6.50; +10.12]
Temporal Duration	300	+7.51	[+5.92; +9.11]
Style	176	+6.31	[+4.82; +7.84]
Update	268	+3.24	[+2.12; +4.39]
Skill	169	+2.32	[+1.04; +3.73]
Multi-hop	249	+1.63	[+0.59; +2.67]
Title	196	+1.55	[+0.85; +2.30]
Proactivity	427	+1.13	[+0.77; +1.52]
Constraint	402	+0.30	[−0.07; +0.69]

Table 9: Events—Raw, paired difference in evidence precision over the nine official EverMemBench categories, percent and pp. The sign is positive everywhere; the CI excludes zero in eight categories out of nine. Compare with Table 8: the category with the largest delivery gain (Single-hop) shows a drop in accuracy, because accuracy there is at the ceiling.

C Exact storage specification

Unit	Fields and purpose
RawTurn	ID, conversation network, speaker, timestamp, text; immutable address.
MemoryEvent	ID, index text, type, viewpoint owner, subject, temporal mode, observed time, evidence, local context.
EventEvidence	Event ID, raw source ID, literal quote; verifiable link.
EpisodeEvent	Episode ID, Event ID, order, attach operation; append-only history.
SearchDocument	kind, index text, render text, source IDs; common retrieval interface.

Table 10: The logical schema does not depend on a JSONL, SQLite or PostgreSQL backend.

D Artifacts and the boundaries of the claims

Code, protocols and configurations: <https://github.com/yaroslavlikh/temporal-episodes> (the repository is closed until publication). What it redistributes follows the licence of each benchmark: per-question

predictions and judgments for EverMemBench (data under Apache 2.0) and SocialMemBench (CC BY 4.0), and for GroupMemBench, which publishes no licence, only aggregates and SHA-256 checksums of the withheld per-question files. Model caches, embeddings and the raw corpora are not redistributed. The identifiers and hashes of the sealed runs are listed below.

The main immutable directories:

- evermembench_temporal_episodes_official_v1_20260915_021550_MSK;
- final_sprint_A_events_full_v1_20260915_233142_MSK;
- final_sprint_B2_episodes_chrono_v1_20260915_232910_MSK;
- final_sprint_C_hyde_raw_v1_20260916_024144_MSK;
- final_sprint_C2_hyde_token_matched_v1_20260916_024306_MSK;
- socialmembench_temporal_episodes_official_harness_v1_20260912_122626_MSK.

An additional sealed GroupMemBench amendment is stored in `final_sprint_D_groupmembench_repair_v2`: 1,490 rows, `results_v2` SHA-256 41ae844eabb292f9d0c7edbeed0476981155340ac85c5ef197d2184fcbd0d4a5, `predictions_v2` SHA-256 20f2c60180a58d37c63f96904896266c8ac5805f83d822d321a34926850c8a96. The directory was not copied into a separate frozen container; its eight source files were verified against the manifest and are unchanged. Revisions used: EverMemBench code e10b3d52f0e4, data a6b210a32248; GroupMemBench e2682e01ff49. The figure script hashes the input summaries/reports. Publishing the code does not by itself grant permission to redistribute the benchmark corpora.

Which Social run is canonical. Two SocialMemBench pipelines are present in the artifacts, and they must not be mixed. The canonical one is `socialmembench_temporal_episodes_official_harness_v1`: text-embedding-3-small, one cosine top-10 per condition, an identical code path for answer/judge/packing; every number in this work comes from it, and its RAW/FLAT/VERSIONED rows are the unmodified results of the frozen run of 2026-09-11. The second, `socialmembench_temporal_episodes_full_v1`, is an earlier hybrid (local MiniLM, raw top-12 + episode top-12, a 400-word budget); its Raw gives an evidence recall of 48.5% against 67.5% for the canonical one. Comparing a condition from one run against Raw from the other produces plausible and entirely false numbers; we note this here because both directories are distributed together.

Recomputation of the evidence endpoint. All the paired numbers of §7.1 and of Table 9 are reproduced offline by the script `research/evidence_endpoint_analysis.py`, which reads only sealed runs, makes no API calls and writes `research/figures/evidence_endpoint_results.json`; the manuscript takes its numbers from there. In the public repository the same values, together with the registered frontier check of §7.3, are recomputed from the published per-question files by `analysis/verify_evidence_endpoint.py`, which compares each one against the value printed here and fails if any of them disagrees.

Admissible formulations. One may speak of the competitiveness of the full system in an adapter evaluation, of a confirmed improvement in *evidence delivery* over the whole dataset and its replication on SocialMemBench, of dominance over matched HyDE in the density of delivered evidence, of a nominal temporal signal in accuracy, and of provenance as addressability of the source. One may not call the result an official SOTA, a proven universal gain, semantically verified facts, or a proven benefit of episode linking. Nor may one claim that events win *because* they improve the recall-precision frontier of delivered evidence: the registered check did not confirm this (§7.3). The confirmed delivery result may not be carried over to answer quality: that is exactly the implication this work tests and does not confirm. An improvement in precision may not be called semantic faithfulness of the derived text, and it may not be extended to GroupMemBench, where evidence metrics were not computed.